

Cambridge (CIE) IGCSE Additional Maths



Your notes

Binomial Theorem

Contents

* Binomial Theorem



Your notes

Binomial Theorem

Binomial Expansion

What is the Binomial Expansion?

- The **binomial theorem** (also known as the binomial expansion) gives a method for expanding a **two-term** expression in a bracket raised to a power
- To expand a bracket with a two-term expression in:
 - First choose the most appropriate parts of the expression to assign to a and b
 - Then use the formula for the binomial theorem:

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n$$

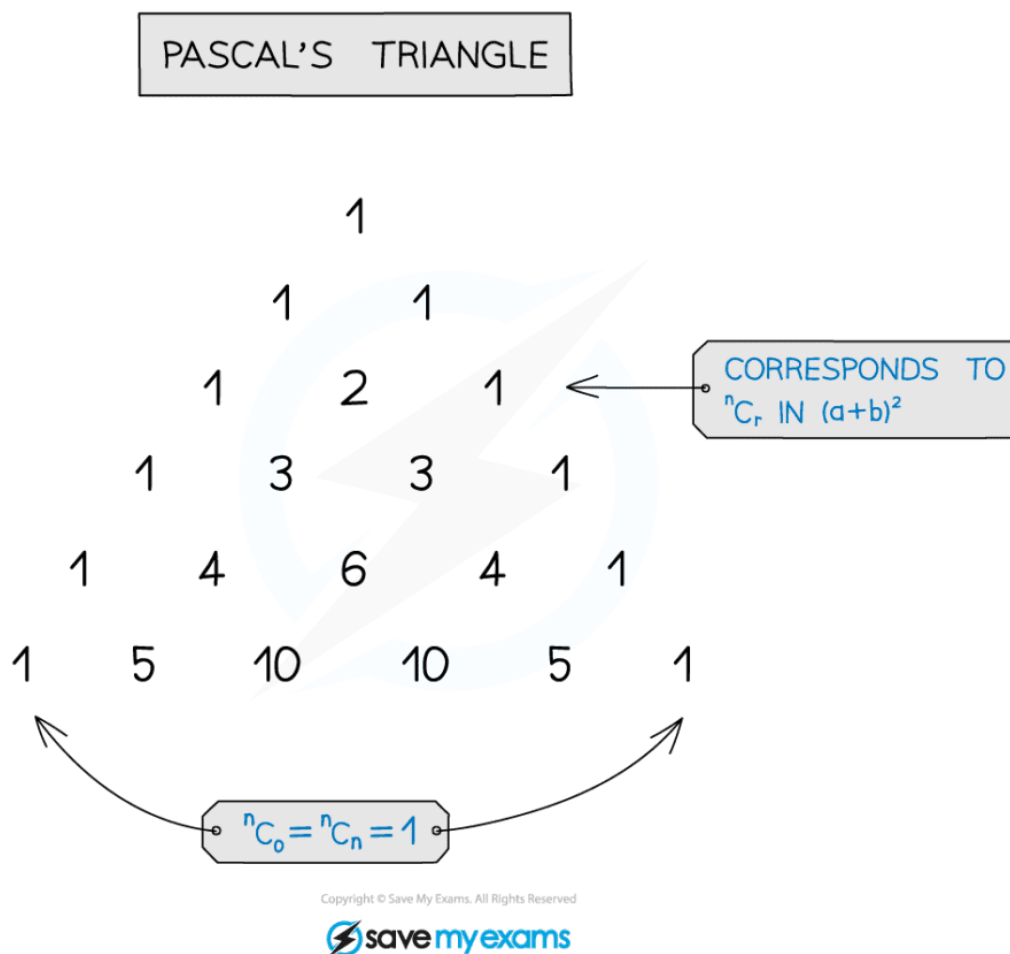
- where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$
 - You may also see $\binom{n}{r}$ written as nC_r or ${}_nC_r$
- You will usually be asked to find the first three or four terms of an expansion

What is Pascal's Triangle?

- Pascal's triangle is a way of arranging the binomial coefficients and neatly shows how they are formed
 - Each term is formed by adding the two terms above it
 - The first row has just the number 1
 - Each row begins and ends with a number 1
 - From the third row the terms in between the 1s are the sum of the two terms above it



Your notes



How does Pascal's Triangle relate to the binomial expansion?

- Pascal's triangle is an alternative way of finding the binomial coefficients, $\binom{n}{r}$
 - It can be useful for finding for smaller values of n without a calculator
 - However for larger values of n it is slow and prone to arithmetic errors
- Taking the first row as the 0th row, $\binom{0}{0} = 1$, each row corresponds to the n^{th} row and the term within that row corresponds to the r^{th} term



Your notes

- The first term, which will be a 1, will be the 0th term
- In the non-calculator exam Pascal's triangle can be helpful if you need to get the coefficients of an expansion quickly, provided the value of n is not too big

How do I find the coefficient of a single term?

- Most of the time you will be asked to find the coefficient of a term, rather than carry out the whole expansion
- Use the formula for the general term

$$\binom{n}{r} a^{n-r} b^r$$

- The question will give you the power of x of the term you are looking for
 - Use this to choose which value of r you will need to use in the formula
 - This will depend on where the x is in the bracket
 - The laws of indices can help you decide which value of r to use:
 - For $(a + bx)^n$ to find the coefficient of x^r use $a^{n-r} (bx)^r$
 - For $(a + bx^2)^n$ to find the coefficient of x^r use $a^{n-\frac{r}{2}} (bx^2)^{\frac{r}{2}}$
 - For $(a + \frac{b}{x})^n$ look at how the powers will cancel out to decide which value of r to use
- So for $(3x + \frac{2}{x})^8$ to find the coefficient of x^2 use the term with $r = 3$ and to find the constant term use the term with $r = 4$
 - There are a lot of variations of this so it is usually easier to see this by inspection of the powers
- You may also be given the coefficient of a particular term and asked to find an unknown in the brackets
 - Use the laws of indices to choose the correct term and then use the binomial theorem formula to form and solve an equation



Examiner Tips and Tricks



Worked Example

Find the first three terms in the expansion of $(3 - 2x)^5$, in ascending powers of x . Simplify the coefficient of each term.

Identify the values of a , b and n .

$$a = 3, b = -2x, n = 5$$

Substitute values into the formula for $(a + b)^n$. The question asks for ascending powers of x , so start with the constant term, a^n .

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n.$$

Pay attention to the negative term for b .

$$\begin{aligned} (3 - 2x)^5 &= 3^5 + \binom{5}{1} (3)^{5-1} (-2x) + \binom{5}{2} (3)^{5-2} (-2x)^2 + \dots \\ &= 243 + (5)(81)(-2x) + (10)(27)(4x^2) \\ &= 243 + (405)(-2x) + 270(4x^2) \\ (3 - 2x)^5 &= 243 - 810x + 1080x^2 + \dots \end{aligned}$$

Applications of Binomial Expansion

What are binomial expansions used for?

- Binomial expansions are used to expand brackets
 - Normally in the form $(a + b)^n$
- You will most likely be asked to find the first few terms
- Look out for whether you should give your answer in **ascending** or **descending** powers of x
 - For **ascending** powers start with the constant term, a^n
 - For **descending** powers start with the term with x in



Your notes

- You may wish to swap a and b over so that you can follow the general formula given in the formula book
- If you are not writing the full expansion you can either
 - show that the sequence continues by putting an ellipsis (...) after your final term
 - or show that the terms you have found are an approximation of the full sequence by using the sign for approximately equals to ($\approx$)

e.g. FIND, IN ASCENDING POWERS OF x ,
THE FIRST THREE TERMS OF $(5+2x)^9$

$$(5+2x)^9 \approx (5)^9 + 9(5)^8(2x) + {}^9C_2(5)^7(2x)^2$$

$$= 1\,953\,125 + 7\,031\,250x + 11\,250\,000x^2$$

APPROXIMATELY EQUAL
TO AS 3 TERMS ONLY

ASCENDING POWERS
OF x SO START
WITH CONSTANT
TERM (x^0)

Copyright © Save My Exams. All Rights Reserved



- You may be asked to find the coefficient of a particular term

e.g. FIND THE COEFFICIENT OF x^6 IN THE
EXPANSION OF $(3+2x)^8$

$${}^8C_6(3)^2(2x)^6$$

$$= 16\,128x^6$$

MATCHES
POWER
OF " b "

START HERE, FOR " x^6 "
" $(b)^6$ " IS NEEDED

TOTAL OF POWERS IS " n "
WHICH HERE IS 8

SO THE COEFFICIENT OF x^6 IS 16 128

Copyright © Save My Exams. All Rights Reserved



- You may be asked to solve a problem to find an unknown



Your notes

e.g. THE COEFFICIENT OF THE x^3 TERM IN THE EXPANSION OF $(p+2x)^5$ IS 2000.
FIND THE VALUE OF p , GIVEN $p > 0$

ONLY NEED THE
TERM IN x^3

$${}^5C_3 (p)^2 (2x)^3 = 80p^2 x^3$$

$$80p^2 = 2000$$

$$p^2 = 25$$

$$p = \cancel{-5} \quad p = 5$$

SAY WHY THIS
SOLUTION IS
REJECTED

REJECT SINCE $p > 0$

$$p = 5$$

Copyright © Save My Exams. All Rights Reserved

